ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Teaching and answer guidance · Road geometry and acceptance

Volume 4 · F1, F2, F3, G1, G2, G3, G4, G5, G6

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In this volume

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Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for road geometry and acceptance. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

Drainage, grade and base

Worked example · Follow the water

A 120 m access road has a sound-looking base. A blocked catch basin sits at its low point, and water is pumping through one wheel path. The paving crew is ready.

Hold the affected work and identify the drainage and support problems. Protect public access and the inlet, document the location, and request the responsible person's correction and acceptance. Do not treat extra asphalt as a drainage repair. Record the repaired outlet and accepted foundation before release.

Draw the four typical layers from prepared soil to asphalt

Draw the four typical layers from prepared soil to asphalt. Label one check at each interface.

Answer guidance.

From bottom: subgrade—support/moisture; sub-base—approved material and accepted lift; base—grade/crossfall/cleanliness; asphalt—specified mixture and compacted thickness. Actual designs may omit or add layers; use the project section.

Observe: Reading, Writing, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

An approved 80 m × 4 m sub-base lift is 0.15 m compacted thickness

An approved 80 m × 4 m sub-base lift is 0.15 m compacted thickness. Find its compacted volume. Can you order that many loose cubic metres without further information?

Answer guidance.

80 × 4 × 0.15 = 48 m³ compacted. Loose supply volume requires a justified conversion or project allowance; 48 m³ loose is not automatically enough.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

A partner proposes opening a ditch toward a nearby creek to remove ponded water

A partner proposes opening a ditch toward a nearby creek to remove ponded water. Give a respectful response and a next step.

Answer guidance.

Explain that drainage changes need the approved design and environmental controls. Hold the change, notify the responsible person and obtain an authorized solution that protects the watercourse and neighbouring land.

Observe: Communication, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify actual layer boundaries on an approved section.	0 / 1 / 2	Evidence/help: _____
Trace a drainage route from surface to authorized outlet.	0 / 1 / 2	Evidence/help: _____
Record and communicate a foundation defect before covering it.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Verify slopes, crowns and grades

Worked example · A level-book entry

Benchmark EL 100.000 m; backsight 1.250 m; foresight to prepared base 1.670 m. Target base EL 99.650 m.

HI = 101.250 m. Existing base = 101.250 − 1.670 = 99.580 m. Target minus existing = +0.070 m: fill 70 mm at this point, subject to the specified material and compaction method. Check against the correct drawing revision and benchmark before directing work.

Use the same benchmark with backsight 1.400 m and foresight 1.600 m

Use the same benchmark with backsight 1.400 m and foresight 1.600 m. Target EL is 99.650 m. State cut or fill.

Answer guidance.

HI 101.400 m; existing EL 99.800 m; target − existing = −0.150 m, so cut 150 mm at the measured point.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

A profile uses horizontal 1:500 and vertical 1:50

A profile uses horizontal 1:500 and vertical 1:50. Explain why measuring its apparent angle is unreliable.

Answer guidance.

The vertical scale is exaggerated ten times relative to horizontal. Calculate actual rise over horizontal run using dimensions and elevations. A screenshot may also be resized.

Observe: Digital, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Write a handover for a disturbed offset stake at station 0+080, right side.

Write a handover for a disturbed offset stake at station 0+080, right side.

Answer guidance.

Identify station, side, stake and observed disturbance; mark the reference as unverified, notify the grade/survey lead and request replacement or verification. Do not continue from an estimated mark.

Observe: Writing, Communication, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Set up and check spirit, builder's and laser levels under supervision.	0 / 1 / 2	Evidence/help: _____
Verify automatic slope control against an independent approved reference.	0 / 1 / 2	Evidence/help: _____
Maintain specified grade and slope and retain a legible level-book record.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Final grade preparation

Worked example · Width before tonnage

A rectangular paving area is 60 m long. The approved width is 4.2 m, but the marked limits measure 4.0 m.

Approved area = 252 m²; marked area = 240 m². The shortfall is 12 m². Verify the drawing and correct the layout before ordering or placing to the wrong boundary. Extra mix later does not automatically restore the intended edge support and joint quality.

The target casting EL is 102.340 m and its verified current EL is 102.295 m

The target casting EL is 102.340 m and its verified current EL is 102.295 m. Find the required elevation change.

Answer guidance.

Raise 0.045 m, or 45 mm, using the approved adjustment system and recheck. This arithmetic does not specify which ring or bedding detail is acceptable.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Build a five-item release record for the base.

Build a five-item release record for the base.

Answer guidance.

Include drawing/revision and location; grade/crossfall checks; compaction/support acceptance; clean surface and drainage; utilities/edges/access ready. Identify checker, date and unresolved holds.

Observe: Writing, Digital through the learner's actual contribution, calculation, explanation or saved work.

A worker suggests smoothing a pumping area and paving immediately to keep the delivery slot

A worker suggests smoothing a pumping area and paving immediately to keep the delivery slot. Respond.

Answer guidance.

Describe the observed pumping, protect the area and request evaluation and correction. Coordinate delivery changes with the crew; schedule pressure does not establish foundation acceptance.

Observe: Communication, Adaptability, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Check utility elevations and protected openings with the responsible person.	0 / 1 / 2	Evidence/help: _____
Verify crown, crossfall and width against the current layout.	0 / 1 / 2	Evidence/help: _____
Complete a release record identifying unresolved defects.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Gradation

Worked example · One good result is not full acceptance

A fictional 600 t lot qualifies for a +$0.75/t gradation adjustment under the complete applicable table conditions. A separate density result is under review.

The isolated gradation adjustment is 600 × $0.75 = +$450. Record it as provisional for this parameter. Do not declare the lot accepted or payable until all applicable requirements and the authorized disposition are resolved.

An audit finds sample Q17 on a report but no matching location log

An audit finds sample Q17 on a report but no matching location log. Write a finding and action.

Answer guidance.

Finding: Q17 cannot be traced to a location/lot from the available log. Preserve records, notify the quality lead, investigate original evidence and obtain an authorized disposition; do not invent a location.

Observe: Writing, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

For a fictional 420 t lot, calculate a −$1.50/t isolated adjustment.

For a fictional 420 t lot, calculate a −$1.50/t isolated adjustment.

Answer guidance.

420 × −$1.50 = −$630. It is a deduction for the identified parameter only, subject to the actual contract and acceptance process.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

Compare the 2006 document and the current project document in a digital register

Compare the 2006 document and the current project document in a digital register. What fields prevent accidental substitution?

Answer guidance.

Record title, issuer, edition/date, amendments, contract adoption, clause/table, file location and approval status. Flag superseded teaching examples separately from project instructions.

Observe: Digital, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Trace one fictional lot through plan, sample, result and disposition.	0 / 1 / 2	Evidence/help: _____
Identify the governing edition and amendments.	0 / 1 / 2	Evidence/help: _____
Communicate a quality hold with exact location and evidence.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Oil content

Worked example · Read the basis before the band

Approved binder content is 5.20 kg/100 kg dry aggregate. The three fictional sub-lot results are 5.45, 5.50 and 5.49 on that same basis. Lot mass is 500 t.

Mean = (5.45 + 5.50 + 5.49)/3 = 5.48. Absolute deviation = 0.28. Under the dated Table 502-F exercise, 0.16–0.30 gives $0/t, so this isolated adjustment is $0. This says nothing about other acceptance parameters.

Approved content is 5.20 and the mean is 4.86, both kg/100 kg dry aggregate

Approved content is 5.20 and the mean is 4.86, both kg/100 kg dry aggregate. Find deviation and the historical isolated adjustment for 300 t.

Answer guidance.

Absolute deviation = 0.34. Table 502-F band 0.31–0.35 gives −$1/t; 300 t gives −$300. Check lift, reporting and all other contract conditions before disposition.

Observe: Numeracy, Reading through the learner’s actual contribution, calculation, explanation or saved work.

A report says 5.2% but has no basis

A report says 5.2% but has no basis. Can it be compared directly with 5.2 kg/100 kg dry aggregate?

Answer guidance.

No. Obtain the test report’s basis and method. A percentage of total mixture is not the same as a percentage of dry aggregate.

Observe: Problem solving, Communication through the learner’s actual contribution, calculation, explanation or saved work.

Write a two-line hold note for conflicting sample IDs.

Write a two-line hold note for conflicting sample IDs.

Answer guidance.

Identify lot, lift, sample IDs and conflicting records; preserve both. Request the quality lead's reconciliation and disposition before changing production or releasing the affected work.

Observe: Writing, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify binder basis and lot/lift before calculation.	0 / 1 / 2	Evidence/help: _____
Read the complete applicable payment/rejection table.	0 / 1 / 2	Evidence/help: _____
Escalate a conflicting result without altering original evidence.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Application

Worked example · Mass over area

A fictional lot receives 288 t over 2,400 m². Specified application is 120 kg/m².

288 t = 288,000 kg. Actual = 288,000/2,400 = 120 kg/m². Ratio = 120/120 × 100 = 100%. In the dated 96.0–103.9% band, isolated adjustment = 288 × $0.50 = +$144, subject to all applicable conditions.

A 200 m × 3.5 m rectangle receives 77 t

A 200 m × 3.5 m rectangle receives 77 t. Find actual application rate.

Answer guidance.

Area = 700 m². Mass = 77,000 kg. Rate = 110 kg/m². Compare with the specified rate only after checking the lot geometry and ticket allocation.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

Specified rate is 115 kg/m²

Specified rate is 115 kg/m². What percentage of specified is 110 kg/m²?

Answer guidance.

110/115 × 100 = 95.652…%, approximately 95.65% before applying the contract's reporting rule. Do not round prematurely to 96% to choose a different band.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Design a ticket register that a partner can audit.

Design a ticket register that a partner can audit.

Answer guidance.

Include ticket ID, truck, date/time, mass/unit, mix, lane/station, lift, lot, disposition of returned material and checker. Use unique IDs and preserve corrections.

Observe: Digital, Writing, Collaboration, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Measure and document a lot boundary.	0 / 1 / 2	Evidence/help: _____
Reconcile unique tickets to the correct lift.	0 / 1 / 2	Evidence/help: _____
Calculate rate and explain why average rate does not prove uniform thickness.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Segregation

Worked example · Describe before diagnosing

A coarse-looking strip appears near the right edge after three truck changes. The surrounding surface has more fine matrix.

Record the repeated location and timing, photograph from a safe position with a scale and location label, and notify the paving/quality lead. Compare with agreed criteria and inspect the process under authorized conditions. Do not declare severe segregation or choose a repair from appearance alone.

Write three useful observations and one unsupported conclusion for the case.

Write three useful observations and one unsupported conclusion for the case.

Answer guidance.

Useful: right-edge location; repeated after three truck changes; reduced visible fine matrix relative to surroundings. Unsupported: “the driver caused it” without investigation.

Observe: Writing, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

In the dated Table 502-I-2, a 1 lane-km top lift has 4 slight areas, zero moderate and zero severe

In the dated Table 502-I-2, a 1 lane-km top lift has 4 slight areas, zero moderate and zero severe. What is the isolated adjustment?

Answer guidance.

The row requiring 4–5 slight, 0 moderate and 0 severe gives +$500 per lane-km. All three columns and the clause conditions must apply. This does not remove remediation obligations.

Observe: Numeracy, Reading through the learner’s actual contribution, calculation, explanation or saved work.

Propose a safe digital defect log and a respectful crew discussion.

Propose a safe digital defect log and a respectful crew discussion.

Answer guidance.

Use location, time, lift, load, image, observed texture and status; avoid personal blame. Ask which process changes occurred and agree who will investigate and report the authorized correction.

Observe: Digital, Communication, Collaboration, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Compare texture using approved project criteria.	0 / 1 / 2	Evidence/help: _____
Locate and record a suspected segregated area.	0 / 1 / 2	Evidence/help: _____
Separate observation, possible cause and authorized disposition.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Density

Worked example · Two legitimate references, two different percentages

In-place density is 2.340 Mg/m³. A fictional plant briquette result is 2.400 Mg/m³; maximum theoretical density is 2.500 Mg/m³.

Marshall-style comparison: 2.340/2.400 × 100 = 97.50%. Maximum-theoretical comparison: 2.340/2.500 × 100 = 93.60%. Both calculations are arithmetic examples; use only the reference required for the actual mixture and contract.

For 2.325 Mg/m³ in place and 2.500 Mg/m³ maximum theoretical, calculate percent.

For 2.325 Mg/m³ in place and 2.500 Mg/m³ maximum theoretical, calculate percent.

Answer guidance.

2.325/2.500 × 100 = 93.00%. Identify the reference as maximum theoretical density and retain the actual method and lot details.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

A 400 t fictional Superpave lot average is 93.4%

A 400 t fictional Superpave lot average is 93.4%. Find the isolated dated table adjustment, then name two checks still needed.

Answer guidance.

400 × $0.50 = +$200. Still check sub-lot/rejection requirements and all other acceptance parameters; verify the correct edition, reference and valid test results.

Observe: Numeracy, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

A spreadsheet divides every test by a fixed 2.400

A spreadsheet divides every test by a fixed 2.400. What should a reviewer do?

Answer guidance.

Check mixture, lot and required reference; replace the unsupported constant with a traceable approved reference for each applicable calculation. Preserve the original record and document the correction.

Observe: Digital, Collaboration, Writing through the learner's actual contribution, calculation, explanation or saved work.

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Explain and select the correct density denominator.	0 / 1 / 2	Evidence/help: _____
Trace test results to location and reference.	0 / 1 / 2	Evidence/help: _____
Distinguish a lot average from individual acceptance requirements.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Smoothness

Worked example · Prorating an authorized historical adjustment

For a fictional exercise, the applicable table and acceptance checks establish a +$800 adjustment for a full 1 km lot. The qualifying auxiliary segment is 565 m.

Length factor = 565/1,000 = 0.565. Prorated adjustment = 0.565 × $800 = +$452. The $800 is an explicitly supplied exercise value, not a quoted Table 502-J value or a current project entitlement.

A qualifying segment is 240 m and the supplied full-kilometre adjustment is −$1,200

A qualifying segment is 240 m and the supplied full-kilometre adjustment is −$1,200. Calculate the exercise adjustment.

Answer guidance.

240/1,000 × −$1,200 = −$288, provided the applicable contract permits this prorating and all conditions are satisfied.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

A learner says a good density result proves a smooth road

A learner says a good density result proves a smooth road. Explain the gap.

Answer guidance.

Density describes compaction relative to a reference. Smoothness describes longitudinal profile. Different measurements and acceptance requirements are needed.

Observe: Communication, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Create a handover for a repeated bump near each truck-change location.

Create a handover for a repeated bump near each truck-change location.

Answer guidance.

Record station, lane, lift, time/load sequence and observation; preserve profile/inspection evidence. Ask the paving and quality leads to investigate supply, paver movement, setup and joints without assigning an unverified cause.

Observe: Writing, Digital, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify the profile measurement and applicable contract limits.	0 / 1 / 2	Evidence/help: _____
Separate IRI, localized defects and crossfall.	0 / 1 / 2	Evidence/help: _____
Trace a profile concern to a location and authorized corrective process.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Integrated practice · all nine Skills for Success

Check the grade and the quantity

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional rectangle is 80 m × 3.5 m. Assigned mix mass is 33.6 t and specified application rate is 120 kg/m². Benchmark EL is 100.000 m, backsight 1.300 m, foresight 1.720 m and target base EL 99.650 m. A draft claims “the application rate proves the whole mat is accepted.” No density or smoothness results are supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Evidence and feedback: Use the exact case facts; distinguish observation, assumption and missing authority.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Evidence and feedback: The note preserves verified facts, units and unresolved holds without inventing a release.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Evidence and feedback: Use the worked numerical key below; show the relationship and its practical limits.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Evidence and feedback: Record the product/capabilities, prompt, source facts, saved files and checks. A chat response alone is not a created or verified file. Check formulas and preserve originals.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Evidence and feedback: Separate the observed issue from a guessed cause; preserve evidence and identify the correct decision role.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Evidence and feedback: Observe an actual exchange and confirmation of the essential facts; do not infer communication from a shared file.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Evidence and feedback: Both learners take an identifiable turn; feedback is respectful and tied to the case evidence.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Evidence and feedback: Use the change-response key below; retain the earlier version and do not alter unrelated facts.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Evidence and feedback: The visual/media has accurate labels, source credits and a stated purpose. Inspect generated diagrams/media; do not invent technical settings. Keep a transcript/captions when producing audio/video.

Change card · Use after the first attempt

A corrected drawing changes width to 3.6 m before placement; the mass allocation has not been revised.

Verified case key

Area 280 m²; actual rate 33,600/280 = 120 kg/m², or 100% of specified. HI 101.300; measured EL 99.580; fill 70 mm at that point. Application rate alone does not establish density, smoothness, uniform thickness or acceptance.

Change-response key

New area 288 m². The same 33.6 t would give 116.67 kg/m², about 97.22% of 120. Confirm the authorized quantity/layout change before treating it as the current plan.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person’s actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.